



Insomnia

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Introduction

Insomnia is defined as dissatisfaction with sleep quality or quantity that is associated with 1 or more of the following features: difficulty falling asleep, difficulty staying asleep or early morning awakening without being able to return to sleep.

The diagnosis requires that the sleep disturbance causes distress or impairment in functioning, occurs at least 3 nights per week for at least 3 months and is not substance-related.^[1]

When the sleep disturbance meets the defined criteria but has occurred for less than 3 months, it is diagnosed as short-term insomnia disorder or other specified insomnia disorder.^{[1][2]} This is sometimes still referred to as acute insomnia.

Insomnia can be a symptom, a risk factor or an independent disorder or it can be comorbid with other sleep disorders or psychiatric and/or medical conditions (e.g., sleep apnea, depression, anxiety, pain). It is critical to identify the presence or absence of comorbid conditions that present with insomnia complaints. Ongoing evaluation is required to assess their severity and impact while treating the insomnia complaint.^[3]

Though pharmacotherapy may often be required, behavioural interventions are the foundation for any treatment of insomnia and range from simply addressing sleep hygiene breaches to providing multimodal treatment known as cognitive behavioural therapy for insomnia (CBT-I); see [Nonpharmacologic Choices](#).

Insomnia associated with significant comorbidities (see [Table 3](#)) is usually more difficult to treat and may require an ongoing combination of medication, behavioural therapy and/or a referral to a sleep medicine clinic.

Goals of Therapy

- Improve daytime function, which is the primary goal of any sleep intervention
 - although patients are typically focused on quantity/quality of their sleep, it is more important to establish how their daytime functioning relates to their sleep
- Reduce daytime impairment (e.g., dysphoria, fatigue, decreased alertness)

- Promote subjectively sound and restorative sleep when external (e.g., stress, noise, jet lag) or internal (e.g., pain, anxiety) factors disrupt natural sleep (i.e., making sleep more resilient)
- Potentiate the effectiveness of behavioural interventions for chronic insomnia^[4]

Investigations

- The assessment and diagnosis of insomnia is a layered approach that involves predisposing factors that increase the risk of developing insomnia (e.g., anxious predisposition), precipitating factors (e.g., emotional distress) and perpetuating factors (e.g., learned negative sleep behaviours).^[3]
- A sleep history ([Table 1](#)) is essential to quantify daytime impairment, determine the outcome of previous pharmacologic and nonpharmacologic interventions, and rule out other sleep pathologies and substances that may be impacting sleep.
- Other measures for assessment include completion of a sleep-wake diary for 1–2 weeks ([Table 2](#)).^[5] Ongoing sleep diaries can be useful for intensive behavioural therapy, but care must be taken not to reinforce maladaptive cognitions and to refocus on daytime function instead. Sleep/insomnia questionnaires such as the [Insomnia Severity Index](#) and the [Pittsburgh Sleep Quality Index](#) can also be utilized in conjunction with scales for comorbidities, not in isolation.
- Objective measurements include actigraphy, a measure of sleep parameters via average motor activity over a period of days–weeks using a noninvasive accelerometer most often housed in a smartwatch.^[6] Actigraphy can supplement a sleep diary in an insomnia workup; however, its accuracy may be compromised by medications or other sleep disorders. Some devices may also exaggerate findings, leading to maladaptive cognitive beliefs about sleep in the absence of clinical interpretation. All actigraphy data, including commercial proxies (e.g., FitBit, smartphone sleep sensors), provide a rough estimate of sleep parameters and need to be taken into context and reviewed with a health-care provider.
- Referral for polysomnography is not recommended in the initial stages of routine insomnia assessment, though it is often necessary in patients with risk factors for or symptoms of other sleep disorders (e.g., sleep apnea, periodic limb movements).^[7] It also should be conducted in resistant or chronic cases and in patients on long-term hypnotic therapy. Home polysomnography is an option for uncomplicated sleep apnea and is standard practice outside large, urban centres.^[8] Its limitations in not diagnosing the sleep problem and only confirming cases of moderate-to-severe obstructive sleep apnea should be understood.
- Screen patients with insomnia for other sleep disorders, medication use and comorbidities, particularly those that may cause nocturnal discomfort or pain such as arthritis, menopause and GERD (see [Table 3](#)). Patient-rated scales can be used to help identify potential comorbidities for further treatment focus, a critical part of the insomnia treatment plan. A number of validated screening tools are outlined in [Table 4](#).

Table 1: Sleep History

Temporal aspects of sleep	Times when patient goes to bed, attempts to sleep, wakes up, gets out of bed
Quantitative aspects of sleep	Sleep latency (time to fall asleep); number and duration of awakenings; wakefulness after sleep onset; total sleep time
Qualitative aspects of sleep	Subjective sleep quality, satisfaction
Behavioural/environmental factors	Non-sleep activities in bed (phone, computer, television); environment (temperature, light, sound); bed partners and pets; perceived causes of awakening

Symptoms of other sleep disorders	Obstructive sleep apnea (snoring, breathing pauses); restless legs syndrome (urge to move legs); parasomnias (sleep walking and talking); circadian rhythm disorders (unusual sleep timing)
Daytime causes/consequences of disturbed sleep	Napping; exercise; work and activities; social and family stressors; caffeine, alcohol, tobacco use

Table 2: Sleep Diary

Date							
Daytime							
Caffeine intake (what, when)							
Alcohol intake (what, when)							
Nicotine use (what, when)							
Naps (start time, end time)							
Exercise (what, when)							
Overall daytime functioning, on a scale of 1 to 5 (1 = alert, energetic; 5 = fatigued, sleepy)							
Nighttime (previous night)							

Date							
What time did you go to bed?							
Did you watch television in bed?							
Did you read in bed?							
How long did it take to fall asleep?							
For each time you woke up during the night, record the following: 1. Time you were awake, e.g., 1:10–2:30 a.m. 2. What woke you up, e.g., snoring, gasping for air, bad dreams, pet, noise							
What time did you wake up for the day?							
What time did you get out of bed for the day?							

Date							
How many hours of sleep did you get?							
How did you feel when you woke up this morning, on a scale of 1 to 5?							
(1 = refreshed, alert; 5 = fatigued, sleepy)							

Table 3: Common Comorbidities and Medications Contributing to Insomnia^{[7][10]}

Psychiatric disorders	- Mood disorders - Attention-deficit hyperactivity disorder (ADHD) - Post-traumatic stress disorder (PTSD)	- Anxiety disorders - Alcohol/substance use disorders
Other medical disorders	- Neurologic (stroke, migraine) - Pulmonary (chronic obstructive pulmonary disease [COPD], asthma) - Digestive (gastroesophageal reflux disease [GERD], colitis) - Chronic pain	- Musculoskeletal (arthritis, fibromyalgia) - Endocrine (hypothyroidism, hyperthyroidism) - Cardiovascular (congestive heart failure)
Medications	- Antidepressants - Stimulants - Antihypertensives - Sedatives	- Decongestants and antihistamines - Analgesics - Herbal supplements - Substances of abuse

Table 4: Select Screening Tools for Key Comorbidities in Insomnia

Sleep Disorders	Mental Health Concerns
Obstructive sleep apnea: StopBANG , Berlin Questionnaire	Depression, mood disorders: PHQ-9 , MDQ , Bipolar Spectrum Diagnostic Scale

Sleep Disorders	Mental Health Concerns
Restless legs syndrome: sIRLS	Anxiety disorders: GAD-7
Circadian rhythm disorders: mCTQ	Attention-deficit hyperactivity disorder (ADHD): ASRS
	Post-traumatic stress disorder (PTSD): PC-PTSD
	Substance abuse: TAPS

Therapeutic Choices

For the management of insomnia, see [Figure 1](#).

Nonpharmacologic Choices

- Good sleep hygiene is the cornerstone of any sleep intervention (see [Table 5](#)). Although data indicates that sleep hygiene alone is often not sufficient to treat insomnia disorders, it is incorporated in the first-line intervention for insomnia, **cognitive behavioural therapy for insomnia** (CBT-I).^{[11][12]} CBT-I is a multi-component intervention offered in up to 8 sessions in both individual and group formats (see [Table 6](#) and [Table 7](#)). CBT-I with consistent follow-up sets the stage for successful treatment of insomnia. CBT-I is also effective for insomnia with comorbidities and may help improve both sleep and the comorbid condition.^[13] Improvements are observed typically after 3–4 sessions.
- Changing misconceptions about sleep need to be addressed prior to recommending behaviour change. The [dysfunctional beliefs and attitudes about sleep \(DBAS\) tool](#) is brief and useful to assess maladaptive cognitions about sleep before embarking on CBT-I.
- The best studies on CBT-I have utilized manualized, systematic therapy provided by trained clinicians in well-selected groups. However, primary care physicians have utilized modified and simplified versions of CBT-I successfully.^{[14][15]} Barriers to CBT-I include lack of trained specialists, financial cost and, most important, patient acceptability and adherence. More accessible means of CBT-I delivery include digital CBT-I, an abbreviated, 2-session version of CBT-I, and use of CBT-I print materials or mobile apps.
- Internet-delivered CBT has proven as effective as other forms of CBT-I and is likely the most accessible.^[16] Online CBT-I programs include:
 - High cost: [Haleo](#), [Sleepio](#), [Somryst](#)
 - Moderate cost: [CBT for Insomnia](#), [Go! to Sleep](#)
 - Free: [CBT-I Coach](#), [My Sleep Well](#)
- Nonetheless, 1 in 4–5 patients does not respond to CBT-I,^[12] while up to 30% of patients drop out of CBT-I.^[17]
- Aerobic exercise, a useful modifier of stress and dysphoric moods, also promotes deeper and more restful sleep.^[18] Encourage patients with insomnia to increase exercise.
- Mindfulness meditation and bright light therapy have both demonstrated modest improvements in sleep efficacy.^[19] A moderately sized RCT of weighted blankets demonstrated a significant reduction of insomnia severity.^[20]
- Acupuncture has demonstrated efficacy in multiple studies, but the evidence is of low quality. The quality of evidence for a variety of other nonpharmacologic strategies (e.g., music, foot baths, sound masking) is very low; however, there are limited risks associated with these strategies.

Table 5: Sleep Hygiene Guidelines

1. Keep a regular sleep-wake schedule with a consistent arising time within 60–90 minutes, 7 days per week. This enhances the circadian sleep rhythm and prevents oversleeping.
2. Restrict time in bed to the average sleep time you have obtained each night over the preceding week.
3. Avoid sleeping in, extensive periods of horizontal rest or daytime napping of more than 15–20 minutes in the afternoon. These activities usually worsen the subsequent night's sleep.
4. Get regular exercise every day: about 30 minutes of an activity with sufficient intensity to cause sweating. If evening exercise prevents sleep, schedule the exercise earlier in the day.
5. Avoid caffeine after mid-afternoon, alcohol anytime after supper, and recreational drugs, all of which disturb sleep. If you must smoke, do not do so after 7:00 p.m.
6. Plan a quiet period before lights out; a warm bath may be helpful. Keep the temperature in your room between 16–20 °C, as this appears to signal the circadian rhythm to sleep.
7. Avoid large meals late in the evening; a light carbohydrate snack (e.g., crackers and warm milk) before bedtime can be helpful.
8. Turn the clock face away. Looking at the clock time on awakening can cause emotional arousal (performance anxiety or anger), delaying a rapid return to sleep. Always use an alarm.
9. As much as possible, keep the bedroom dark and soundproof. Avoid looking at your phone, television or computer in the bedroom. Electronic devices emit light and can be alerting depending on the activity they are being used for. If you live in a noisy area, consider ear plugs. If light is difficult to control, use an eye mask.
10. Use the bedroom only for sleep and intimacy; using the bed as a reading place, office or media centre conditions you to be alert in a place that should be associated with quiet and sleep. If you awaken during the night and are wide awake, get up, leave the bedroom and do something quiet until you feel drowsy/tired, then return to bed.
- Note:** Pharmacologic (or any) interventions will be less effective if these guidelines are not followed. In mild cases of insomnia, sleep hygiene strategies, practised consistently and together, may be sufficient to reinstate a normal sleep pattern.

Table 6: 5 Components of CBT-[21][22]

Component	Description
1. Stimulus Control	- Go to bed only when sleepy - Get out of bed after 15 minutes if sleep does not come - See Table 5
2. Sleep Hygiene	See Table 5

Component	Description
3. Sleep Restriction (paradoxical approach)	- Limit time in bed to actual time asleep or 6–7 hours - Increase time in bed by 20–30 minutes per week if sleep increasing, or decrease this if sleep still fragmented - Usually best done under supervision of health-care provider
4. Relaxation Techniques	Prevent hyperarousal through meditation, progressive muscle relaxation, autogenic training
5. Cognitive Therapy	Target and change dysfunctional beliefs/attitudes about insomnia (see Table 7)

Table 7: Targeted Cognitive Barriers to Sleep[23]

Advice for Cognitive Barrier	Elaboration
Keep realistic expectations	Even good sleepers do not always get 8 hours of quality sleep. There are individual differences in sleep requirements.
Do not blame insomnia for all daytime impairments	Consider alternative explanations that might contribute to those impairments.
Do not catastrophize after a poor night's sleep	Insomnia can be unpleasant, but it is not dangerous. The worst outcome: you will be sleepier the next day and sleep more soundly the next night.
Do not give too much importance to sleep	Even if sleep occupies one-third of your life, don't make it the sole focus of your existence.
Develop some tolerance to the effects of insomnia	Rearrange your schedule, but do not cancel planned activities.

Pharmacologic Choices

Before considering pharmacological treatment of insomnia, the following should be implemented and documented: sleep education, discussion of CBT-I strategies, diagnostic sleep studies (if necessary) and optimization of treatment of major comorbidities. Evidence suggests that hypnotic medications are comparably efficacious to CBT-I during acute treatment.[24] Combination of pharmacotherapy with CBT-I has proven effective and should be considered.[25][26]

Many medications with a Health Canada–approved indication for the treatment of insomnia have significant side effects and are potentially problematic when used long-term (see [Table 10](#)). Due to stringent exclusion criteria, these agents were studied in populations without major comorbidities and may not be representative of up to 75% of patients with insomnia who have a major comorbid condition.[27][28] Accordingly, it may be reasonable to use sedating medications off-label to promote sleep for comorbid insomnia if the medication is an effective choice for that particular comorbidity.

Use the lowest effective dose for the shortest duration. Although the limitation of a 7–10 day up to a maximum of 1 month supply with hypnotic agents is arbitrarily determined, potentially damaging and not evidence-based, frequent follow-up (3–6 weeks) and limited dispensing (30–60 days) is warranted, especially at the beginning of insomnia treatment.

If sleep performance and daytime functioning have not returned to normal after 2–6 weeks, re-evaluate the diagnosis and comorbidities with consideration of referral to a sleep specialist. The results of a longitudinal study suggest that up to two-thirds of patients meet the criteria for chronic insomnia.^[29] Thus, long-term sedative use is often seen in practice. Although long-term sedative use should be the exception rather than the rule, it may be required in cases with significant comorbidities and/or failure of or inability to complete nonpharmacologic therapies. A clear rationale, ongoing assessment and clinical awareness of risks/benefits of long-term sedative use is essential.^[30] Some hypnotics have demonstrated safety and efficacy for up to 12 months of use (eszopiclone,^[31] zolpidem,^[32] lemborexant^[33]).

Benzodiazepines

All benzodiazepines have sedative and hypnotic properties, but differ significantly in potency and pharmacokinetics. Benzodiazepines also confer a significant risk of adverse effects, such as delirium, falls, motor vehicle accidents, respiratory depression, cognitive impairment, and memory issues as well as abuse, dependence, and withdrawal symptoms with long-term use.^[34] This is of particular concern in older patients or those who are medically ill.^[35] Although some studies have suggested benzodiazepines confer an increased risk of mortality and dementia, this has likely been overestimated due to study design; newer data suggest no clear association.^{[36][37]} Risks may be attenuated by using for a short duration and, possibly, by intermittent use.^[38]

In Canada, 4 benzodiazepines (**flurazepam**, **nitrazepam**, **temazepam**, **triazolam**) are officially indicated for insomnia. Temazepam is usually a suitable all-purpose hypnotic with a half-life sufficient to cover the sleep period without causing hangover effects.^[30] Flurazepam and nitrazepam are not recommended due to their longer half-lives; they accumulate with repeated dosing and are associated with more next-day effects than shorter-acting agents. In older patients, they cause higher cortical impairment, resulting in confusion and falls.^[39] The use of triazolam is also not recommended because its pharmacokinetic properties may confer a higher risk of abuse, dependence and rebound insomnia. It also has a unique, dose-related adverse effect profile (confusion, agitation, amnesia).^[40]

Benzodiazepines should be considered for short-term acute or intermittent use. Long-term use should be considered only in severe or comorbid insomnia cases that have failed other treatments and when the benzodiazepine has demonstrated a substantial improvement in functional outcome (not only sedation). Benzodiazepines for the treatment of insomnia have been studied for up to 24 weeks, and rebound was observed upon discontinuation.^[41] Deprescribing strategies, very gradual tapering (i.e., over a period of months), lowering dosage or substitute treatments/prescriptions should always be reviewed with the appropriate patients. However, aggressively restricting, tapering or stopping stable low doses of benzodiazepines in an indiscriminate fashion may lead to more damaging behaviours such as increased unmonitored OTC medication or substance misuse.^[42] A harm-reduction approach must be realized for long-term benzodiazepine use, as the resources to taper/discontinue may not be present or the benzodiazepine may improve the quality of life in those particular cases due to significant insomnia comorbidity and/or treatment resistance.

Benzodiazepine Receptor Agonists (Z-Drugs)

Although not benzodiazepines, the Z-drugs, **zopiclone**, **eszopiclone** and **zolpidem**, act at the benzodiazepine receptor and have similar therapeutic effects on overall sleep, but more limited effects on sleep structure and fewer adverse effects, including cognitive impairment, dependence, tolerance and rebound insomnia.^{[31][32][43][44][45]} They have 10–40 times less of a myorelaxant effect than benzodiazepines^[46] and they have not demonstrated worsening of sleep apnea.^[47] These differences may be attributed to the relative affinity of the Z-drugs to the various subtypes of the GABA-A alpha receptor (see [Table 8](#)). Although the Z-drugs are associated with less risks and clearly belong to a different drug class, much of the same guidance described for the use of benzodiazepines can be applied to the Z-drugs (e.g., lowest dose for shortest duration).

Though noted as rare, complex sleep behaviours have been reported in up to 3% of patients prescribed zopiclone and zolpidem.^[48] This potential adverse effect should be reviewed with all patients considering Z-drugs, including how it can be partially mitigated by ensuring that the medication is taken in bed with limited access to potentially dangerous situations (e.g., kitchen tools, car keys). Combination with alcohol has been shown to increase all adverse events including complex sleep behaviours; this contraindication should be repeatedly stressed.^[49]

Zopiclone, a cyclopyrrolone, is the most commonly prescribed Z-drug in Canada. **Eszopiclone** is the active (S+) isomer of zopiclone; it shares similar pharmacologic properties to the racemic compound but has a different GABA-A binding subunit profile (see [Table 8](#)) and may be more potent, have quicker onset and longer duration of action with less next-day effects than zopiclone.^[50] Although both zopiclone and eszopiclone are recommended to be taken at least 12 hours before driving or operating machinery, zopiclone appears to confer a greater actual risk of impaired driving.^{[51][52]} Eszopiclone may offer greater efficacy and tolerability than other agents excluding zopiclone since it was not included in a large meta-analysis.^[53] A Cochrane systematic review supports the efficacy of eszopiclone for transient, chronic and comorbid insomnia with no consistent evidence of tolerance and withdrawal.^[54]

Zolpidem is an imidazopyridine with specific affinity to the alpha-1 subtype of the GABA-A alpha subunit.^[55] It has a shorter half-life than zopiclone and eszopiclone and requires only 8 hours before operating machinery or driving. Zolpidem is considered a controlled substance, similar to benzodiazepines, but unlike zopiclone and eszopiclone.

Zaleplon is a pyrazolopyrimidine that also has specific affinity to the alpha-1 subtype, but with an even shorter half-life than zolpidem. It has been discontinued commercially, but is often available from compounding pharmacies. The normal dosage range is 5–10 mg up to a maximum of 20 mg. It can be very useful for patients requiring treatment for initial insomnia with limited next-day adverse effects.^[56]

Table 8: Binding Affinity of Z-drugs and Benzodiazepines^{[44][57]}

Distribution and Functional Effect of GABA-A Receptor Alpha Subunits		Affinity for Receptor			
Subunit	Putative Role	Benzodiazepines	Zopiclone	Eszopiclone	Zolpidem
Alpha 1	Hypnotic, anticonvulsant, memory	High	Moderate-high	Moderate-high	High
Alpha 2	EEG activity, anxiolytic	High	Moderate	High	Low-moderate
Alpha 3	Sleep, anxiolytic, antidepressant	High	Low-moderate	High	Low-moderate
Alpha 5	Learning, memory	High	Moderate	Low	Very low

Dual Orexin Antagonists

Orexins are neuropeptides that promote a state of wakefulness by stimulating orexin–1 and –2 receptors. Competitive dual orexin antagonists (DORAs) normalize sleep-wake function by reducing wakefulness and unwanted transitions between wake and sleep.^[61] **Lemborexant** and **daridorexant** are DORAs indicated for the treatment of both sleep onset and sleep maintenance insomnia. There is no evidence of withdrawal symptoms or rebound insomnia upon withdrawal, and they have efficacy data for up to 12 months.^{[33][62][63]} Somnolence is a reported adverse effect, but there appear to be minimal effects

on the next day.^{[63][64]} They are not controlled substances, and abuse potential appears minimal. A period of 7 hours is recommended before driving or operating machinery. Although there are no direct comparisons, data suggests that lemborexant may be more effective, whereas daridorexant more tolerable.^[65]

Other Hypnotics

Doxepin, a tricyclic antidepressant (TCA), acts as a selective histamine H1 receptor antagonist at very low dosages (3–6 mg). Therefore, doxepin exerts its sedative effect by antagonism of the histaminic portion of the wake system, as opposed to promoting sleep by activating the gabaergic system.^[58] It is indicated by Health Canada for sleep-maintenance difficulties with positive trials up to 3 months in length^[59] and may improve sleep maintenance more than GABA-A receptor agonists.^[60] Doxepin is not associated with rebound or dependence, has minimal next-day side effects and does not have typical TCA side effects due to the low dosage.

Though still available, chloral hydrate and barbiturates are not recommended for the treatment of insomnia, given their toxicity, potential for abuse and the availability of several safer medications.

Off-Label Agents Used for Insomnia

For more information on off-label medications used for the treatment of insomnia, see [Table 9](#).

When insomnia is comorbid with another condition, rational use of off-label agents may be appropriate, despite a lack of evidence (e.g., sedating antidepressants for depressive disorders, quetiapine for bipolar disorder, gabapentin or pregabalin for fibromyalgia, TCAs for migraine).^{[66][67]} Eszopiclone, zolpidem and lemborexant are agents with official indications for insomnia that may also be effective for insomnia with comorbidities.^{[54][55][64]} Conversely, the use of Health Canada–approved agents may be relatively contraindicated in other comorbidities (e.g., alcohol use disorder for Z-drugs and benzodiazepines), making off-label treatments more prominent in clinical decision-making. In practice, when an off-label agent is used for the treatment of comorbid insomnia, appropriate education, consent, monitoring and documentation should be completed.

Of note, trazodone use for insomnia without comorbidities is widespread, with apparent success and support by some experts in the field.^{[30][68]} Actual data to support this practice is limited, consisting of a few small studies of low-dose trazodone (25–100 mg).^{[69][70]}

Table 9: Off-Label Agents Commonly Used for Comorbid Insomnia^{[30][66][67][68][69][70]}

Drug Class	Reasons for Use	Considerations
Sedating antidepressants: - Mirtazapine (low-dose)^[71] - Tricyclic antidepressants (including doxepin >10 mg) - Trazodone	- Insomnia with mood disorder^[72] - mirtazapine may be useful for low appetite or alcohol-use issues - Comorbid migraine and other central sensitivity syndromes (TCAs only)	- Next-day sedation and motor restlessness can occur - TCAs also associated with anticholinergic adverse effects - Both TCAs and mirtazapine may be associated with weight gain - Trazodone may better for sleep maintenance, not overall efficiency
Antihistamines: Chlorpheniramine	Insomnia associated with histamine-mediated sleep disturbance (e.g., allergies, atopic dermatitis)	Excessive risk of daytime sedation, psychomotor/cognitive impairment and

Drug Class	Reasons for Use	Considerations
- Diphenhydramine - Hydrazine	- Very short term - Rarely recommended due to side effect profile	anticholinergic toxicity, especially in older patients
Anticonvulsants: - Gabapentin, low-dose - Pregabalin, low-dose	Insomnia associated with centralized pain syndromes (e.g., fibromyalgia, neuropathic pain, restless legs syndrome)	- Weight gain and next-day sedation are common adverse effects - CNS depression and cognitive impairment may also occur
First-generation antipsychotics - Chlorpromazine, low-dose - Methotrimeprazine - Loxapine	Insomnia associated with very resistant bipolar disorder or schizophrenia	Not recommended for insomnia in the absence of comorbidities, due to unacceptable risk of anticholinergic and neurological toxicity
Second-generation antipsychotics - Olanzapine - Quetiapine - Paliperidone - Clozapine - Brexpiprazole - Risperidone - Lurasidone - Asenapine	- Insomnia with bipolar disorder or schizophrenia (olanzapine, quetiapine, clozapine paliperidone, risperidone, asenapine, lurasidone)^[73] - Insomnia with major depressive disorder (augmentation with brexpiprazole)^[74] - Insomnia with generalized anxiety disorder (quetiapine)^[75]	- Not recommended for insomnia in the absence of comorbidities, due to unacceptable risk of metabolic syndrome - Metabolic issues and weight gain must be monitored
Other benzodiazepines - Diazepam - Clonazepam - Lorazepam - Nitrazepam - Alprazolam - Oxazepam	- Insomnia with anxiety disorders or severe hyperarousal - Insomnia associated with restless legs syndrome (clinically done, minimal evidence to support)	

Natural Supplements and Over-the-Counter Agents Used for Insomnia

The use of natural supplements and/or over-the-counter agents is common; a Canadian study found that over 14% of patients reported using them. ^[76] Of the natural supplements, **melatonin** has the most evidence, with a small overall effect size;^{[77][78]} **valerian** and **L-tryptophan** have inconsistent results but are occasionally clinically useful.^{[79][80]} See [Insomnia](#) in *Compendium of Therapeutics for Minor Ailments* for more detailed information on natural supplements.

Screen patients for use of over-the-counter sleep agents that contain **diphenhydramine**, **dimenhydrinate** or **doxylamine** (e.g., Tylenol PM, Benadryl, Gravol, NyQuil Cold & Flu). The use of these agents is highly discouraged, due to increased risk of cognitive impairment, hangover effects, dizziness, anticholinergic delirium and falls, especially in older adults.^{[81][82]}

Given the legalization and subsequent heavy marketing of **cannabis** products for sleep, cannabis use should also be reviewed with patients. There is insufficient evidence to support routine clinical use of cannabis for the treatment of insomnia; the effects of the many components of cannabis on sleep remains unclear. Cannabis should be considered only in resistant cases of insomnia with a comorbidity that has some evidence for its efficacy (e.g., pain syndromes, resistant anxiety) and after failure of treatments with better evidence.^[83] Cannabis products consist of mainly 2 ingredients—cannabidiol (CBD) and tetrahydrocannabinol (THC). Although THC is more sedating, it is psychoactive in nature; therefore, CBD-dominant products are usually preferred initially.

Special Populations

Insomnia and Children

Nonpharmacologic therapy is recommended as first-line treatment of insomnia in children, especially young children (<6 y).^[84] Graded extinction (i.e., gradual withdrawal of parental attention to child crying over sleeping alone) is an additional behavioural technique that is often useful in younger children and infants. Use of medication for pediatric insomnia is not evidence-based and is guided largely by empirical data, clinical experience and treatment of comorbidities.

If pharmacologic therapy is required, melatonin may be considered as an option (initial dose of 0.5–1 mg at bedtime).^[85] A reasonably sized meta-analysis demonstrated that melatonin was an effective and tolerable drug for the short-term treatment of sleep onset insomnia in children and adolescents,^[86] particularly in children with ADHD and autism spectrum disorder.^[87] Significant adverse effects have not been observed in studies with a follow-up period of up to 7 years.^[88]

Other natural products have limited evidence, but may be considered due to minimal risk.^[89] For example, **L-tryptophan** has demonstrated potential efficacy in children with parasomnias.^[90] Over-the-counter antihistamines are not recommended due to risk of rapid tolerance, next-day sedation, cognitive impairment and paradoxical reaction.^[91]

Prescription medications should be prescribed extremely judiciously and targeted at the child's comorbidity if necessary.

Clonidine has demonstrated efficacy in a case-series of children with both ADHD and insomnia.^[89] Z-drugs have not demonstrated efficacy in children,^[87] while benzodiazepines and TCAs are not recommended due to an unfavorable benefit:risk ratio.^[91]

Insomnia and Older Patients

Older patients are prone to polypharmacy and potential for drug interactions, and typical pharmacological treatments for insomnia are further limited due to the greater potential of deleterious side effects such as cognitive impairment and falls. Thus, the administration of CBT-I as first-line therapy is even more paramount in this population.

Safer options for pharmacotherapy include lower-dose doxepin (<6 mg) for sleep maintenance or melatonin for sleep-onset insomnia.^{[43][92]} Lemborexant also demonstrated a favourable 6-month safety and efficacy profile in older patients.^[93]

In general, benzodiazepines and Z-drugs should be avoided in older patients due to increased risk of falls, confusion and increased potency.^[34] Of note, these warnings are based on retrospective studies with numerous confounding variables. Of these agents, zolpidem or zaleplon (from compounding pharmacies) may be preferred due to their shorter half-life.^[43] Low-dose eszopiclone (2 mg) has also demonstrated short-term efficacy (up to 24 wk) with a low risk of falls.^{[94][95]}

Low-dose trazodone and mirtazapine are commonly used but have very little evidence in older patients. They should be reserved for insomnia with comorbid depression/anxiety or more resistant cases due to conflicting evidence and potential for falls and fractures.^{[96][97]}

Although deprescribing of hypnotic therapy should be discussed and offered, consider the risks and benefits of long-term, low-dose therapy in stable and functional older patients. Overt rigidity about deprescribing may lead to adverse outcomes in patients who have been functioning favourably for years, as they may replace the hypnotic with other harmful substances (e.g., OTC drugs with a high anticholinergic burden, cannabis or alcohol). Keep in mind that off-label psychotropics for insomnia and indicated drugs for other medical conditions (e.g., antihypertensives, antidepressants, antipsychotics) also come with their own fall risks.^[98]

Insomnia and Pregnancy

Disrupted sleep is one of the most frequently reported complaints of pregnant patients and it tends to worsen as the pregnancy progresses.^[99] Multiple endogenous factors, such as endocrine changes, physical discomfort and bladder distention, contribute to this common complaint. In the postpartum period, exogenous factors, e.g., the child's sleep-wake and feeding schedules, further disrupt sleep. Additionally, psychological factors, e.g., anxiety or the emergence of psychiatric disorders during the pregnancy or after the birth, can add to the insomnia burden with persistent sleep complaints increasing the risk of postpartum mood disorders.^[100] Two sleep disorders, sleep apnea^[101] and restless legs syndrome,^[102] are known to worsen during pregnancy and may present as insomnia in the pregnant patient (see [Restless Legs Syndrome](#)).

There has been little research on the effects of insomnia during pregnancy on maternal functioning and there are no randomized controlled studies of any intervention in this population. Regardless of cause, it is likely that significantly disrupted sleep during pregnancy and in the postpartum period will impair the patient's quality of life by causing similar daytime impairments to those seen in the nonpregnant population;^[103] this impairment may adversely affect the parent-child bond.^[100]

Management of Insomnia during Pregnancy and Postpartum Period

Although not well studied, clinical experience suggests that cognitive and behavioural techniques are effective in pregnant patients with insomnia and are the treatment of choice when distress and impairment are mild to moderate.^[104] Screening and treatment of insomnia may prevent stress-related perinatal complications,^[105] as pregnancy may be an ideal time to integrate CBT-I strategies for improved overall health.^[106]

If insomnia is severe or disabling, carefully consider the risk-benefit ratio to both the patient and the developing fetus when using a pharmacologic intervention.^[107] Although lacking empirical evidence, intermittent dosing using the safest medication at the lowest effective dose, combined with behavioural strategies, is reasonable.

Zopiclone, eszopiclone and trazodone do not appear to be teratogenic;^{[107][108]} they may be reasonable choices if clinically justified. Benzodiazepines should be avoided, particularly in the first trimester, due to an increased risk of oral cleft; they may also cause neonatal withdrawal symptoms when used closer to term.^[109]

Although doxepin does not appear to be teratogenic, it has not been studied in pregnancy. Antidepressants also do not appear to be teratogenic, but there is an extremely low risk of respiratory problems and pulmonary hypertension in the newborn; much of these studies are confounded with the underlying mood disorder.^[110] Zolpidem is not recommended in pregnancy as it has been associated with a higher risk of preterm delivery, small-for-gestational-age newborns and caesarean delivery but no increased risk of congenital anomalies.^[111]

Since the safety of products containing lemborexant, melatonin, L-tryptophan and diphenhydramine in pregnancy is unknown, they are not recommended. Melatonin may pose a theoretical disruption of reproduction, and should likely be avoided in conception and pregnancy.^[107]

Insomnia and Breastfeeding

If pharmacotherapy is required during breastfeeding, intermittent low doses of short-acting **benzodiazepines** such as temazepam or **Z-drugs** appear relatively safe with appropriate monitoring; however, the effects of chronic exposure are unknown.^{[107][112]}

The safety of most other hypnotics in breastfeeding is unknown. Although **trazodone** is transferred to breast milk in low amounts, it is not expected to affect breastfed infants, especially those >2 months of age and when doses ≤100 mg/night are used.^[113] Doxepin and diphenhydramine should not be used in breastfeeding patients, based on observed adverse effects in case reports.^[107]

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

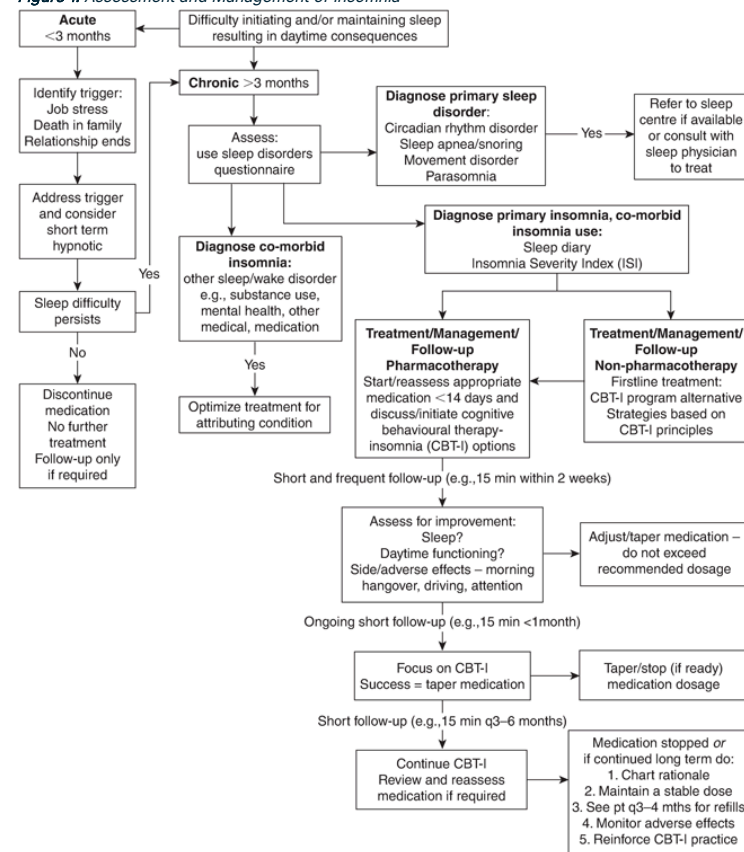
Therapeutic Tips

- If risk factors or symptoms dictate, ensure that patients with insomnia undergo a sleep study to rule out sleep apnea. The major comorbidities that must be reviewed initially are depression, anxiety, sleep apnea and chronic pain.
- Educate the patient about sleep and sleep hygiene and set realistic treatment goals to minimize daytime impairment rather than specifically increase sleep performance; a chronic poor sleeper will not be turned into a good sleeper overnight; realistic expectations are essential for a successful treatment plan. Behavioural therapy may take 3–4 weeks to demonstrate improved sleep, whereas pharmacotherapy may provide some relief in the first few days.
- The degree of daytime impairment directs the intervention: if there is an acute change in daytime functioning, a short course of hypnotics may be indicated; if the daytime impairment is mild or chronic, try a behavioural intervention (e.g., sleep restriction) first.
- There is a marked dichotomy between theory and practice regarding hypnotics. Long-term hypnotic use may be necessary for select patients and should not be completely dismissed. Reluctance to allow patients access to long-term hypnotics may cause unnecessary suffering and lead to unmonitored self-medication with more deleterious substances.
- Warn patients about combined effects when hypnotics are used with other CNS depressants. They should never be combined with alcohol.
- The recommended time frame after hypnotic administration and before driving or operating machinery should be followed initially, but keep in mind that sleep deprivation itself can cause significant issues with alertness.^[51] Habituation to driving effects has been shown with regular use of hypnotics;^[114] off-label sedatives have not been tested rigorously and may actually be more deleterious. Use clinical judgment and consider the nature of the patient's driving (more vigilance required for commercial drivers), provincial requirements in reporting unfit drivers, and balancing next-day side effects of hypnotics with functional improvement of insomnia.
- Plan to taper and withdraw hypnotics at a low-stress time (e.g., vacation, not during an acute stressor). An extremely slow tapering strategy can also be helpful. If complete withdrawal is not achievable, a lower dose or one used less frequently is also beneficial and may improve patient motivation and alliance. A rigid dogma of requiring complete discontinuation of the hypnotic is not usually helpful in establishing this.

- Insomnia often occurs as a symptom, comorbid condition or prodrome of other psychiatric conditions such as depression and/or anxiety disorders.^{[115][116]} Chronic sleep disturbance of at least 1 year in duration increases the risk of a mood disorder in subsequent years.^{[117][118]} Vigilance is required for the emergence of mood and anxiety disorders, which should be managed with appropriate specific therapy. Optimization and adjustment of diagnosis of mood and anxiety disorders should be considered concurrently with, or possibly prior to, sleep treatment. Screen for other comorbidities before, after and during insomnia treatment, especially if the patient has a reduced response to therapy.

Algorithms

Figure 1: Assessment and Management of Insomnia



Modified with permission. Toward Optimized Practice (TOP) Insomnia Group. 2015 December. Assessment to management of adult insomnia: clinical practice guideline. Edmonton AB: Toward Optimized Practice. Available from: actt.albertadoctors.org/CPGs/Pages/Adult-Insomnia.aspx

Drug Table

Table 10: Health Canada–Approved Medications Used to Manage Insomnia

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Benzodiazepines				
temazepam Restoril < \$5	Initial: 15 mg QHS PO Maximum: 30 mg QHS PO Maximum in older patients (≥65 y): 15 mg QHS PO	Dose-dependent ataxia, dizziness; dependence/withdrawal symptoms, impaired memory, risk of abuse. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake.	Substrate of CYP3A4; metabolism could be increased by enzymes inducers or decreased by enzyme inhibitors. Dosage adjustments may be necessary. Do not combine with alcohol or other CNS depressants.	Benzodiazepines are not recommended in older patients or in those with respiratory disorders or OSA. Significant rebound symptoms reported after discontinuation. An off-label benzodiazepine (e.g., lorazepam, clonazepam) may be considered if lower dose required. Use only when there is a period of at least 7–8 h before planned awakening. Advise patients to wait ≥12 h before driving or operating machinery.
Drug Class: Benzodiazepine Receptor Agonists (Z-drugs)				

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
eszopiclone  Lunesta, generics \$20–35	Initial: 1 mg QHS PO	All Z-drugs: bitter/metallic taste, dry mouth, dizziness and somnolence. Complex sleep behaviours such as night eating and somnambulism with no recollection of such activities. Discontinue immediately if such reactions occur. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake. Benzodiazepine class effects seen, but likely less common. Dose-dependent ataxia, dizziness; dependence/withdrawal symptoms, impaired memory, risk of abuse.	Substrate of CYP3A4; metabolism could be increased by enzymes inducers or decreased by enzyme inhibitors. Dosage adjustments may be necessary. Do not combine with alcohol or other CNS depressants. Increased risk of complex sleep behaviours in combination with other CNS -active drugs. Coadministration with food results in slower absorption and possible reduced efficacy.	May have a faster onset of action than zopiclone and less next-day impairment. ^[54] Significant evidence for treating insomnia with a wide variety of comorbidities. Minimal rebound and tolerance. Some evidence to support use in older patients for up to 24 wk. ^{[97][98]} Newly available in Canada (2020), although it has been available internationally for >10 y. Use only when there is a period of at least 7–8 h before planned awakening. Advise patients to wait ≥12 h before driving or operating machinery.
	Maximum: 3 mg QHS PO			
	Maximum in older patients(≥65 y), severe hepatic or renal impairment or coadministered with potent CYP3A4 inhibitors: 2 mg QHS PO			
zolpidem Sublinox, generics \$5–20	Men: 5–10 mg QHS PO	All Z-drugs: bitter/metallic taste, dry mouth, dizziness and somnolence. Complex sleep behaviours such as night eating and somnambulism with no recollection of such activities. Discontinue immediately if such reactions occur. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake. Benzodiazepine class	Substrate of CYP3A4; metabolism could be increased by enzymes inducers or decreased by enzyme inhibitors. Dosage adjustments may be necessary. Do not combine with alcohol or other CNS depressants. Increased risk of complex sleep behaviours in combination with	Avoid in patients with a history of somnambulism or complex sleep behaviours. May be useful for insomnia with comorbidities. ^[55] Minimal rebound and tolerance. May work more rapidly than eszopiclone/zopiclone. Much lower incidence of bitter/metallic taste. Use only when there is a period of at least 7–8 h before planned awakening.
	Women: 5 mg QHS PO			
	Maximum: 10 mg QHS PO Maximum in older patients(≥65 y), mild-moderate hepatic impairment: 5 mg QHS PO			

Drug/Cost ^①	Dosage	Adverse Effects	Drug Interactions	Comments
		effects seen, but likely less common. Dose-dependent ataxia, dizziness; dependence/withdrawal symptoms, impaired memory, risk of abuse.	other CNS -active drugs. Coadministration with food results in slower absorption and possible reduced efficacy.	Advise patients to wait ≥12 h before driving or operating machinery.
zopiclone  Imovane, generics	Initial: 3.75 mg QHS PO Maximum: 7.5 mg QHS PO	All Z-drugs: bitter/metallic taste, dry mouth, dizziness and somnolence. Complex sleep behaviours such as night eating and somnambulism with no recollection of such activities. Discontinue immediately if such reactions occur. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake. Benzodiazepine class effects seen, but likely less common. Dose-dependent ataxia, dizziness; dependence/withdrawal symptoms, impaired memory, risk of abuse.	Substrate of CYP3A4; metabolism could be increased by enzymes inducers or decreased by enzyme inhibitors. Dosage adjustments may be necessary. Do not combine with alcohol or other CNS depressants. Increased risk of complex sleep behaviours in combination with other CNS -active drugs.	Does not accumulate; may cause less rebound on withdrawal. Use only when there is a period of at least 7–8 h before planned awakening. Advise patients to wait ≥12 h before driving or operating machinery.
< \$5	Maximum in older patients(≥65 y), mild-moderate hepatic impairment, renal impairment or coadministered with potent CYP3A4 inhibitors: 5 mg QHS PO			

Drug Class: Dual Orexin Receptor Antagonists (DORA)

Drug/Cost ^①	Dosage	Adverse Effects	Drug Interactions	Comments
lemborexant Dayvigo	Initial: 5 mg QHS PO Maximum: 10 mg QHS PO	Somnolence; less commonly: sleep paralysis, hypnagogic/hypnopompic hallucinations, cataplexy-like symptoms. Complex sleep behaviours such as night eating and somnambulism with no recollection of such activities. Discontinue immediately if such reactions occur. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake.	Not recommended in combination with moderate-strong CYP3A inhibitors or inducers, alcohol or other CNS depressants. Increased risk of complex sleep behaviours in combination with other CNS -active drugs. Coadministration with food results in slower absorption and possible reduced efficacy.	No overt potential for misuse. Contraindicated in narcolepsy. Use only when there is a period of at least 7–8 h before planned awakening. Limited effects on driving impairment after 9 hours.
\$20–35	Maximum in moderate hepatic impairment or coadministered with weak CYP3A inhibitors: 5 mg QHS PO			

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
daridorexant Quviviq	50 mg QHS PO	Headache, nasopharyngitis, somnolence, dizziness, nausea, fatigue; less commonly: sleep paralysis, hypnagogic/hypnopompic hallucinations, cataplexy-like symptoms. Complex sleep behaviours such as night eating and somnambulism with no recollection of such activities. Discontinue immediately if such reactions occur. May cause dose-dependent, next-day impairment of activities requiring alertness, including driving a car, despite the patient feeling fully awake.	Contraindicated in combination with strong CYP3A4 inhibitors; dosage adjustments required when used with moderate inhibitors. Not recommended in combination with CYP3A4 inducers, alcohol or other CNS depressants. Increased risk of complex sleep behaviours in combination with other CNS -active drugs.	No overt potential for misuse. Contraindicated in narcolepsy. Use only when there is a period of at least 7–8 h before planned awakening.
\$20–35	Maximum in moderate hepatic impairment or coadministered with moderate CYP3A4 inhibitors: 25 mg QHS PO		Coadministration with food results in slower absorption and possible reduced efficacy.	

Drug Class: Tricyclic Antidepressants

doxepin Silenor	3–6 mg 30 min before bedtime PO	Somnolence, sedation, nausea.	Metabolized by CYP2D6 and CYP2C19; inhibitors of these enzymes may increase exposure to/toxicity of doxepin; dosage adjustments may be necessary. Not recommended in combination with alcohol or other CNS depressants. Should not be taken within 3 h of a meal to minimize	Recommended treatment for older patients. Limited next-day impairment, rebound or potential for abuse. Adverse effects related to TCA activity possible at doses >10 mg.
\$5–20	In older patients (≥65 y) or hepatically impaired: start with 3 mg QHS PO; may increase to 6 mg if needed			

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
				drowsiness the next day.
^[a] Cost of 14-day supply; includes drug cost only.  Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment. Abbreviations: CNS = central nervous system; MAOI = monoamine oxidase inhibitor; OSA = obstructive sleep apnea; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor				

Suggested Readings

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